

Welcome to the world of Causal Inference!

Etienne Dagorn

September 4, 2026

Who Am I?

- Etienne Dagorn, Assistant Professor in Economics at LEM-U Lille.
- Research interests: education economics, experimental economics, and causal inference.
 - Mail: etienne.dagorn@univ-lille.fr
 - Office hours: by appointment ;

This course

- **Organisation:** 8 lessons of 3 hours.
 - Monday morning: 9:00–12:00.
- **Content:** causal inference and policy evaluation methods and applications.
- **Prerequisites:** basic econometrics.
- Information available on my website:
<https://etiennedagorn.github.io/edagorn/>
- The course website contains slides, homework, and short R exercises.

Expectations and Evaluation

Weekly rhythm

About one hour of exercises between classes, mixing lecture material, readings, and short applied work in R. Main points are corrected at the beginning of the next lecture.

Component	Format
QCM quizzes, 40%	One short in-class online quiz per session; best 6 of 8 retained
Final exam, 60%	In-class written exam on applied policy-evaluation scenarios

Evaluation focus

The goal is not to memorise methods, but to justify comparisons, assumptions, estimands, and policy interpretations.

Structure of the class

1. Foundations of Policy Evaluation
2. Selection Bias, Regression, Matching, and Inference
3. Randomised Experiments: Design and Power
4. Compliance, IV, LATE, and External Validity
5. Difference-in-Differences: 2x2 and TWFE
6. Event Studies and Staggered DiD
7. Regression Discontinuity
8. Synthetic Control, Heterogeneity, and Replication

Recurring reflex

Where does the identifying variation come from? What assumption makes the comparison causal? What estimand or population does the estimate describe?